

# NinjaCart: Transforming Fresh Produce Supply Chains in India

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*A comprehensive case study on India's leading B2B fresh produce supply chain platform, analyzing market inefficiencies, business model architecture, gap analysis, and strategic options for sustainable growth.*

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## Abstract

This comprehensive case study examines NinjaCart, India's leading B2B fresh produce supply chain platform addressing the critical challenges of perishable goods distribution. Unlike farm-input platforms (e.g., DeHaat, AgroStar) or output-focused marketplaces, NinjaCart targets the structural inefficiencies in moving fruits and vegetables from farm to fork through an asset-light, full-stack model combining sourcing, logistics, quality control, and retail distribution. The study is structured into three interconnected sections. First, a detailed market study analyzes India's Rs.6,00,000 crore perishable produce market, examining structural inefficiencies (30%+ post-harvest waste), farmer pain points (price discovery, payment delays), and competitive dynamics. Second, a deep dissection of NinjaCart's business model explores its procurement-to-delivery flywheel, revenue architecture (commission, logistics fees, data insights), unit economics, and dark store network. Third, a systematic gap analysis identifies product portfolio limitations (lack of farmer credit), service delivery shortcomings (tier-2/3 city coverage), technology integration deficits (predictive demand for hyperlocal inventory), and strategic vulnerabilities relative to competitors like WayCool, Jumbotail, and Otipy. The case concludes with actionable recommendations, financial projections, and exhibits including supply chain diagrams, farmer pain-point analyses, margin comparisons, growth trajectory charts, and competitive positioning matrices.

# 1 Introduction: The Perishable Supply Chain Imperative

India's fresh produce sector occupies a paradoxical position in the nation's economy. On one hand, India is the world's second-largest producer of fruits and vegetables (320 million metric tonnes annually), employing nearly 50 million farming households. On the other, productivity and post-harvest management remain stubbornly low compared to global benchmarks. One of the most critical yet underappreciated drivers of this productivity gap is the dysfunctional supply chain for perishable produce—fruits, vegetables, and staples.

An estimated **30-35% of fresh produce is wasted** post-harvest—worth over Rs.2,00,000 crore annually—due to fragmented supply chains, lack of cold storage infrastructure, and multiple intermediary layers. For a smallholder farmer with less than two hectares of land—and there are over 100 million such farmers in India—this waste is not merely a financial loss; it is a threat to household survival.

The average Indian farmer receives only **15-25% of the final consumer price** for fresh produce. Industry estimates suggest that up to 40% of the potential value is lost due to:

- Lack of direct market access (forced to sell to local aggregators at low prices)
- Payment delays of 15-45 days from intermediaries
- No cold storage access (produce wilts within 24-48 hours)
- Price volatility (prices can drop 50% within a single harvest week)
- No quality grading or certification (all produce sold as "mixed" at lower prices)

## 1.1 The Founding Story

NinjaCart was founded in 2015 by Thirukumaran Nagarajan, Kartheeswaran KK, and Sharath Loganathan—alumni of IIT Madras and NIT Trichy with prior experience at Amazon and Cognizant. Their insight was operationally profound: *What if a retailer could order fresh produce via a mobile app before 8 PM and receive quality-assured, graded vegetables at their store by 6 AM the next morning—at 15% lower cost than the wholesale market?*

The name "**NinjaCart**" reflects the company's focus on speed, precision, and silent efficiency in a notoriously chaotic supply chain—moving silently and swiftly like a ninja while delivering value.

## 1.2 Current Status and Scale

Today, NinjaCart serves over **30,000+ retailers** (kirana stores, restaurants, small grocery chains, and institutional buyers) across **15+ cities** (Bengaluru, Chennai, Hyderabad, Mumbai, Delhi NCR, Pune, Ahmedabad, Lucknow, Jaipur, Coimbatore, Vizag, Nagpur, Indore, Surat, and Kochi). The company sources from **50,000+ farmers** across 12 states, operates **45+ dark stores** (fulfillment centers), and generates approximately **Rs.2,500 crore in Gross Merchandise Value (GMV)**. The company has raised over \$200 million in equity funding from investors including Tiger Global, Accel, Syngenta Ventures, Trifecta Capital, and Mirae Asset.

Yet significant gaps remain: limited farmer credit offerings, weak presence in tier-2/3 cities, no hyperlocal demand prediction for dynamic inventory allocation, emerging competition from full-stack agri-platforms, and lack of value-added product lines (cut vegetables, ready-to-cook). This case analyzes those gaps and charts strategic options for the company's leadership.

## 2 Market Study: Anatomy of India's Fresh Produce Supply Chain

### 2.1 Total Addressable Market and Segmental Breakdown

The Indian fresh produce market (fruits + vegetables + staples) is estimated at approximately **Rs.6,00,000 crore** (\$72 billion USD) in 2026, growing at a compound annual growth rate (CAGR) of 8-10%. However, the *addressable organized B2B supply chain market*—where NinjaCart operates—is currently Rs.15,000-20,000 crore, with rapid growth as retail formalizes and kirana stores digitize.

**Table 1** provides a detailed breakdown of the market by segment, including current size, growth rates, margin characteristics, and NinjaCart's current penetration level.

Table 1: Total Addressable Market for B2B Fresh Produce Supply Chain (2026)

| Segment                          | Market (Rs. Cr) | CAGR       | Gross Margin | Penetration   |
|----------------------------------|-----------------|------------|--------------|---------------|
| Fruits (B2B)                     | 1,20,000        | 9%         | 10-15%       | Low           |
| Vegetables (B2B)                 | 1,80,000        | 10%        | 12-18%       | Medium        |
| Staples (onion, potato, tomato)  | 1,50,000        | 7%         | 5-8%         | High          |
| Exotic/heritage produce          | 15,000          | 20%        | 25-30%       | Emerging      |
| Value-added (cut veg, RTC)       | 8,000           | 30%        | 20-25%       | Negligible    |
| <b>Total (B2B fresh produce)</b> | <b>4,73,000</b> | <b>10%</b> | —            | <b>&lt;2%</b> |

#### Key observations:

- Staples remain the largest volume segment but lowest margin (5-8%)
- Exotic/heritage produce and value-added are fastest-growing (20-30% CAGR)
- Vegetables offer the best balance of scale and margin (12-18%)
- NinjaCart has strong penetration only in metro vegetables; significant upside exists in fruits, exotics, value-added, and tier-2/3 cities

### 2.2 The Traditional Supply Chain: Anatomy of Inefficiency

To appreciate NinjaCart's innovation, one must first understand the legacy system it seeks to replace. The traditional fresh produce supply chain in India involves **5-6 intermediary layers**:

|                                                                                                          |
|----------------------------------------------------------------------------------------------------------|
| <b>Farmer</b> → Village Aggregator → Mandi Trader → Wholesaler →<br>Sub-wholesaler → Retailer → Consumer |
|----------------------------------------------------------------------------------------------------------|

#### 2.2.1 Key Inefficiencies

1. **Price discovery failure:** Farmers receive only 15-25% of the final consumer price. The remaining 75-85% is captured by intermediaries, with no transparency on pricing.
2. **Payment delays:** Farmers typically wait 15-45 days for payment from intermediaries. Many are forced to accept lower prices for immediate cash.

3. **No quality standardization:** No grading or sorting at origin. Retailers must accept "as is" produce mixed with damaged or undersized items. There is no recourse for poor quality.
4. **Massive waste:** Lack of cold chain infrastructure means produce loses 20-40% of value within 24-48 hours of harvest. Cold storage exists only for potatoes, onions, and apples—not for green vegetables or most fruits.
5. **No demand signal to farmers:** Farmers plant based on last year's prices, leading to boom-bust cycles (e.g., tomato prices crashing from Rs.100/kg to Rs.5/kg within weeks).
6. **Fragmented transportation:** Small trucks (2-5 tonne capacity) make multiple trips with partial loads, increasing per-unit logistics costs by 40-60% compared to optimized full-truck loads.

### 2.3 Farmer Pain Points: Quantitative and Qualitative Analysis

In 2025, a comprehensive survey of 5,000 farmers across Andhra Pradesh, Karnataka, Maharashtra, Tamil Nadu, and Uttar Pradesh was conducted to identify and rank the most significant pain points in fresh produce marketing. The results are summarized in Table 2.

Table 2: Farmer Pain Point Analysis (N = 5,000, margin of error  $\pm 2\%$ )

| Pain Point                                       | Respondents (%) |
|--------------------------------------------------|-----------------|
| Low price realization (15-25% of consumer price) | 86%             |
| Payment delays (15-45 days)                      | 79%             |
| No price transparency / bargaining power         | 74%             |
| Post-harvest waste due to no cold storage        | 68%             |
| No access to quality certification/grading       | 62%             |
| High transportation costs to mandi               | 55%             |
| Unable to plan planting (no demand forecast)     | 51%             |
| No access to formal credit                       | 47%             |
| No insurance for crop/price risk                 | 42%             |
| Poor quality inputs from local shops             | 38%             |

#### 2.3.1 Analysis of Pain Points

The most severe pain point, reported by **86% of respondents**, is low price realization. Farmers perceive—accurately—that they capture only a small fraction of the final retail price while bearing all production risk.

The second most severe pain point, reported by **79%**, is payment delays. Unlike input purchases (which require upfront cash), produce sales involve long payment cycles. A farmer who sells Rs.50,000 worth of tomatoes may wait 30 days for payment—during which time they cannot purchase inputs for the next crop.

**Post-harvest waste affects 68% of farmers.** A farmer with no cold storage must sell all produce within 24-48 hours of harvest, regardless of price. This urgency is exploited by intermediaries who offer take-it-or-leave-it prices.



**No demand forecasting affects 51% of farmers.** Planting decisions are made based on last year’s prices—a classic cobweb cycle. When many farmers plant the same crop, prices crash at harvest.

## 2.4 Competitive Landscape: Mapping the Rivals

NinjaCart does not operate in a vacuum. A diverse set of competitors, ranging from traditional mandis to emerging agri-tech platforms, vie for the retailer’s fresh produce wallet. Table 3 provides a structured comparison of major competitive models.

Table 3: Competitive Landscape (2026)

| Competitor         | Model          | Focus       | Pricing      | Advantage   |
|--------------------|----------------|-------------|--------------|-------------|
| Traditional Mandi  | Informal       | Local       | Opaque       | Low cost    |
| WayCool Foods      | Full-stack     | South India | Transparent  | Depth       |
| Jumbotail          | Kirana B2B     | Pan-India   | Commission   | Non-produce |
| Otipy              | Group buying   | NCR, Mumbai | Subscription | Low waste   |
| BigBasket          | B2C/B2B hybrid | Metro       | Dynamic      | Brand       |
| Farmers Fresh Zone | Co-op          | Kerala, TN  | Member-based | Trust       |
| Crofarm            | Agri-logistics | Pan-India   | Per-kg       | Asset-light |

## 2.5 Regulatory and Structural Constraints

Any analysis of the fresh produce supply chain must account for the regulatory environment, which imposes significant constraints on business model innovation.

### 2.5.1 Formal Regulations

The **Agricultural Produce Market Committee (APMC) Act**—though reformed in many states—still requires produce to pass through mandis in some jurisdictions. NinjaCart must navigate varying state-level regulations on direct farm procurement.

The **Goods and Services Tax (GST) regime** imposes:

- 5% GST on fresh produce (if processed/packaged)
- 0% on unbranded, loose fresh produce (but no input credit)
- 18% on value-added products (cut vegetables, ready-to-cook)

This cascading structure creates complexity and compliance costs.

The **Essential Commodities Act** historically imposed stock limits on staples (onion, potato, tomato). While amended in 2020, states still invoke stock limits during price spikes, disrupting supply chains.

### 2.5.2 Structural Constraints

**Land fragmentation** is the most significant structural constraint. According to the 2015-16 Agricultural Census, 86% of Indian farmers operate on less than two hectares. Small holdings produce small, inconsistent volumes, making direct sourcing challenging without aggregation.

**Crop perishability** creates extreme logistics pressure. Leafy greens (coriander, spinach, methi) lose value within 12 hours of harvest. Berries and ripe fruits (tomatoes, strawberries) have a shelf life of 3-5 days. This requires time-compressed, temperature-controlled logistics.

**Fragmented retail** means the average order value per retailer is low (Rs.500-1,500). This makes profitable delivery challenging without route optimization and order aggregation.

The **digital divide** remains another barrier. While smartphone penetration among urban retailers is high (70%+), many kirana store owners still prefer phone-based ordering. NinjaCart has addressed this through a call center and WhatsApp-based ordering, but digital adoption for advanced features (predictive ordering, inventory management) remains limited.

### 3 NinjaCart's Business Model: A Deep Dissection

#### 3.1 Foundational Logic and Strategic Positioning

NinjaCart's business model rests on a foundational insight that distinguishes it from both traditional mandis and other B2B platforms: *Retailers don't need produce; they need guaranteed availability of quality produce at predictable prices, delivered before opening hours.*

This shifts the value proposition from commodity trading to service delivery. The retailer is not buying vegetables; the retailer is purchasing a guaranteed outcome: fresh, graded produce available at 6 AM with zero waste and zero stockouts.

This positioning has profound implications for customer acquisition, retention, and lifetime value. Rather than competing solely on price, NinjaCart competes on reliability, quality consistency, and delivery precision. Rather than spending heavily on mass advertising, the company relies on word-of-mouth from satisfied retailers.

#### 3.2 The Procurement-to-Delivery Flywheel

The operational heart of NinjaCart's model is what the company calls the "**procurement-to-delivery flywheel**"—a self-reinforcing loop that begins with reliable sourcing and builds trust through quality consistency.

##### 3.2.1 Flywheel Mechanics Explained

The flywheel operates as follows:

1. A retailer needs fresh vegetables for the next day's sales. Before 8 PM, the retailer opens the NinjaCart app (or calls the toll-free number) and specifies:
  - Products needed (e.g., 5 kg tomatoes, 3 kg onions, 2 kg potatoes)
  - Quality grade required (Grade A - premium, Grade B - standard)
  - Delivery location (store address)
2. NinjaCart's demand-supply matching algorithm aggregates orders from thousands of retailers in each city to determine:
  - Total quantity needed per SKU per delivery zone
  - Quality grade mix
  - Preferred delivery sequencing
3. The aggregated demand is transmitted to:
  - **Farmer clusters** (direct sourcing from 50,000+ farmers) - 70% of volume
  - **Partner aggregators** (pre-vetted village-level collectors) - 20% of volume
  - **Spot mandi purchases** (to fill gaps) - 10% of volume
4. Sourced produce is transported to the nearest **dark store** (fulfillment center) between 10 PM and 2 AM.
5. At the dark store, produce undergoes:

- Quality inspection (visual, texture, spot checks)
  - Grading (Grade A, B, or reject)
  - Weighing and packaging (standardized 500g, 1kg, 5kg units)
  - Cold storage (10-15 minute maximum dwell time)
6. Orders are picked, packed, and loaded onto delivery vans between 3 AM and 5 AM.
  7. Delivery vans depart on optimized routes between 5 AM and 6 AM. Retailers receive their orders by 6:30 AM—before their stores open.
  8. The retailer inspects the order, pays via UPI (immediate) or weekly invoice (for verified repeat customers), and rates the quality.
  9. Positive ratings build trust. The retailer shares the experience with neighboring store owners—critical in retail markets where peer validation is the primary trust signal.
  10. Lower customer acquisition costs (CAC) from referrals free up resources to:
    - Expand the dark store network
    - Improve quality control protocols
    - Add new produce categories
    - Train more farmers on post-harvest handling
  11. These improvements further increase reliability, restarting the loop.

### 3.2.2 Why the Flywheel Works

Several features of this flywheel deserve emphasis:

- **Front-loads value:** The retailer receives quality-assured produce before opening hours, with no upfront payment (for credit customers). This reduces perceived risk and encourages trial.
- **Creates lock-in:** Once a retailer has integrated NinjaCart into their daily opening routine, the transaction cost of switching to another provider—re-learning ordering, quality standards, delivery timing—is high.
- **Data-rich feedback loop:** Every interaction generates data about demand patterns, quality rejection rates, sourcing efficiency, and retailer preferences. This data continuously improves the matching algorithm and inventory planning.

## 3.3 Revenue Architecture: Streams, Margins, Levers

NinjaCart’s revenue model has four primary streams, which have evolved as the company has scaled.

### 3.3.1 Primary Stream: Commission on Produce Sales (70-80% of revenue)

The dominant revenue stream is the commission earned on produce sales facilitated through the platform.

NinjaCart connects farmers and aggregators with retailers, taking a **commission of 12-18%** of the transaction value. The typical retail price for graded vegetables is Rs.30-80 per kg, depending on:

- Product category (exotics command premium over staples)
- Quality grade (Grade A vs Grade B)
- Season (off-season produce 40-60% higher)
- Order volume (bulk orders get volume discounts)

NinjaCart also operates as a principal (taking ownership of inventory) for approximately 40% of volume—primarily for staples and high-volume vegetables. On principal sales, NinjaCart captures the full retail margin (12-18% gross margin) but bears inventory risk (spoilage, price drops).

**Gross margin on commission:** 12-18% (agency model) to 15-20% (principal model during normal conditions). Weighted average gross margin: approximately 14-16%.

### 3.3.2 Secondary Stream: Logistics and Fulfillment Fees (10-15% of revenue)

Retailers pay a separate delivery fee based on:

- Distance from dark store (Rs.10-50 per order)
- Order value (free delivery above Rs.1,000)
- Delivery slot (early morning slots attract premium)

**Gross margins on logistics:** 5-10% (delivery is largely a pass-through cost to third-party logistics partners). Primary value is not profit but enabling the core produce transaction.

### 3.3.3 Tertiary Stream: Data and Insights Subscription (5-10% of revenue)

NinjaCart aggregates anonymized data from millions of retailer interactions and sourcing patterns:

- SKU-level demand forecasts by geography, day of week, season
- Quality rejection rates by source farm/aggregator
- Price elasticity (how demand changes with price)
- Retailer ordering patterns (lead time, frequency, basket composition)

This data is valuable for multiple external stakeholders:

- **FMCG companies** (HUL, Nestlé, ITC) pay for data on retailer reach and shelf dynamics

- **Government agencies** (Ministry of Agriculture, state horticulture boards) pay for data on price trends and supply chain efficiency
- **Food processing companies** pay for data on seasonal availability and pricing

Current annual data revenue: approximately Rs.50-75 crore (2-3% of total revenue). Management expects this to grow to Rs.200 crore by 2028 as data assets accumulate.

### *3.3.4 Quaternary Stream: Supplier Services (2-5% of revenue)*

NinjaCart offers paid services to its farmer and aggregator suppliers:

- **Quality certification** (GlobalGAP, IndiaGAP) - Rs.5,000-15,000 per farm
- **Post-harvest training** (handling, grading, packaging) - Rs.2,000-5,000 per farmer
- **Input access** (seeds, fertilizers, pesticides at discounted rates)
- **Financial products** (invoice factoring, equipment financing) - via NBFC partners

Gross margins on supplier services: 20-30% (training and certification are high-margin). This stream is in early stages.

## **3.4 Unit Economics: Customer Lifetime Value Analysis**

Understanding NinjaCart's unit economics is essential for evaluating the sustainability of its business model. Table 5 presents the key metrics for a representative retailer acquired in 2026.

Table 4: NinjaCart Unit Economics (FY2026 Estimates)

| Metric                                      | Value                     |
|---------------------------------------------|---------------------------|
| Customer Acquisition Cost (CAC)             | Rs.800–1,200 per retailer |
| - Digital advertising                       | Rs.300–500                |
| - Field sales agent commissions             | Rs.200–300                |
| - Referral incentives                       | Rs.100–150                |
| - Promotional offers (first order)          | Rs.200–250                |
| Average Order Value (AOV)                   | Rs.1,200–1,800 (3-5 kg)   |
| Purchase frequency (per week)               | 5-6 days                  |
| - Vegetables                                | 6 days/wk                 |
| - Fruits                                    | 3 days/wk                 |
| - Staples                                   | 2 days/wk                 |
| Gross Profit per Order (GPO)                | Rs.150–250 (12-15%)       |
| <b>Annual Gross Profit per Customer</b>     | <b>Rs.3,900–7,800</b>     |
| 12-month retention rate                     | 65%                       |
| 24-month retention rate                     | 48%                       |
| 36-month retention rate                     | 32%                       |
| <b>3-year Customer Lifetime Value (LTV)</b> | <b>Rs.8,000–12,000</b>    |
| <b>LTV:CAC Ratio</b>                        | <b>8-10:1</b>             |

### 3.4.1 Analysis of Unit Economics

The LTV:CAC ratio of 8-10:1 is healthy by any standard (venture capital benchmarks typically consider 3:1 acceptable and 5:1 strong). This suggests that NinjaCart can profitably invest in customer acquisition.

However, several caveats apply:

1. **Retention is sensitive to quality:** Retailers who experience even one quality failure (e.g., delivered wilted spinach) often churn permanently, regardless of past positive experiences. The cost of a quality failure is not just the refund but the lost lifetime value.
2. **Gross margins face pressure:** As competition increases (especially from Way-Cool and Otipy), NinjaCart may need to reduce commissions or offer discounts to retain retailers.
3. **CAC may increase with expansion:** As NinjaCart expands to tier-2/3 cities without existing word-of-mouth networks, paid acquisition costs will likely rise.
4. **Order frequency varies significantly:** Vegetable orders are daily (high frequency), but fruits and staples are less frequent. New retailer cohorts show lower

frequency before trust is established.

### 3.5 Operations and Fulfillment: The Dark Store Network

The operational backbone of NinjaCart’s model is its network of **dark stores** (fulfillment centers) and last-mile delivery partners.

#### 3.5.1 Dark Store Design

NinjaCart operates **45 dark stores** across 15+ cities. Each dark store serves a **5-10 km radius** covering 500-1,000 retailers. The stores are located in warehouse districts or industrial areas with good road connectivity to both sourcing corridors and retail clusters.

Each dark store is configured as follows:

Table 5: Dark Store Configuration

| Component                | Specification                                                |
|--------------------------|--------------------------------------------------------------|
| Coverage radius          | 5-10 km                                                      |
| Retailers served         | 500-1,000                                                    |
| Daily order volume       | 2,000-5,000 orders                                           |
| Cold storage capacity    | 10-20 tonnes                                                 |
| Sorting/packing stations | 10-15                                                        |
| Delivery vans            | 15-25 (2-tonne capacity)                                     |
| Staff                    | Store manager, 10-15 pickers/packers, 5-7 quality inspectors |

#### 3.5.2 Inventory Management and Dynamic Replenishment

NinjaCart’s inventory management is driven by predictive algorithms that forecast demand based on:

- Historical order patterns (same day of week in previous weeks)
- Weather forecasts (rain predicts lower footfall, lower retailer demand)
- Festival calendar (higher demand for specific produce before festivals)
- Price trends (price elasticity models for each SKU)
- Real-time order data (demand spikes visible 3-4 hours before cutoff)

#### Target freshness metrics:

- Time from harvest to dark store: ¡12 hours for vegetables, ¡24 hours for fruits
- Time from dark store to retailer: ¡2 hours
- Spoilage rate (waste): 3-5% (industry average 15-20%)



### 3.5.3 Last-Mile Delivery Optimization

For last-mile delivery, NinjaCart uses a hybrid model:

- **High-density clusters:** Company-operated delivery vans with optimized routing (using proprietary route optimization algorithm). Target delivery density: 15-20 stops per van per route.
- **Medium-density clusters:** Partner delivery providers (Shadowfax, LoadShare, Delhivery) with SLAs.
- **Low-density clusters:** 3PL couriers for non-time-sensitive orders (staples, packaged goods).

**Target service level for delivery:** 95% of orders delivered by 7 AM; 99% by 8 AM.

## 3.6 Technology Stack: AI, Analytics, and Automation

Underpinning NinjaCart's operations is a proprietary technology stack that integrates ordering, sourcing, inventory, quality control, and delivery.

### 3.6.1 Component 1: Demand Forecasting Engine

The forecasting engine processes order data from multiple channels (mobile app, WhatsApp, website, call center). It considers:

- Retailer historical ordering patterns (per SKU, per day of week)
- Aggregate demand trends (city-level, zone-level)
- External factors (weather, festivals, school holidays)
- Price elasticity models (how demand changes with price)

The engine uses a multi-stage ensemble model (XGBoost + LSTM + Prophet) to generate SKU-level forecasts at 1-hour granularity.

**Current performance:** MAPE (Mean Absolute Percentage Error) of 12-15% at SKU level; 6-8% at category level.

### 3.6.2 Component 2: Quality Control and Grading System

All produce at dark stores undergoes a standardized QC process:

1. **Visual inspection:** Color, size, shape, external damage
2. **Texture check:** Firmness (for tomatoes, fruits), crispness (for leafy greens)
3. **Spot testing:** Random sampling for internal defects (cut testing)
4. **Grading:** Grade A (premium), Grade B (standard), Reject (return to farmer or compost)

NinjaCart is developing a **computer vision-based grading system** using smartphone cameras at sourcing points. Early pilot shows 80% accuracy in detecting external defects, reducing reliance on manual inspection.

### 3.6.3 Component 3: Route Optimization Engine

The route optimization engine plans delivery routes for each dark store considering:

- Order delivery deadlines (by 6 AM, 7 AM, 8 AM tiers)
- Vehicle capacity (weight, volume)
- Traffic patterns (real-time from Google Maps API + historical)
- Retailer delivery windows (some stores prefer 5:30 AM, others 6:30 AM)

The engine uses a modified vehicle routing problem (VRP) solver with time windows, balancing:

- Minimizing total distance traveled
- Minimizing number of vehicles needed
- Maximizing on-time delivery rate

**Current performance:** 92% on-time delivery rate; average route efficiency improvement of 15-20% over manual planning.

### 3.6.4 Component 4: Retailer Relationship Management (RRM) System

The RRM system tracks each retailer's complete interaction history:

- Order patterns (SKUs, quantities, frequency)
- Quality ratings and complaints
- Payment history and credit score
- Store characteristics (location, size, customer demographics)

This longitudinal data enables:

- **Personalized recommendations:** "You bought 5 kg tomatoes last Tuesday; re-order now for 5% off."
- **Credit underwriting:** Retailers with 6+ months of positive history are offered 15-day payment terms.
- **Churn prediction:** Retailers who order less than their historical pattern are flagged for retention campaigns.

## 3.7 Financial Performance: Top-line Growth and Bottom-line Realities

While NinjaCart is a private company and does not publicly disclose detailed financials, industry estimates and investor presentations provide a reasonable picture.

### 3.7.1 Gross Merchandise Value (GMV) Growth

Table 6: GMV Growth Trajectory

| Fiscal Year      | GMV (Rs. Cr) | Growth (%) |
|------------------|--------------|------------|
| FY22             | 650          | —          |
| FY23             | 1,100        | 69%        |
| FY24             | 1,600        | 45%        |
| FY25             | 2,100        | 31%        |
| FY26 (estimated) | 2,500        | 19%        |
| FY27 (projected) | 3,100        | 24%        |
| FY28 (projected) | 4,000        | 29%        |

### 3.7.2 Profitability Profile

Table 7: Profitability Metrics

| Metric                                     | FY24  | FY25  | FY26 (est) | FY28 (target) |
|--------------------------------------------|-------|-------|------------|---------------|
| Gross Margin (%)                           | 11%   | 13%   | 15%        | 18%           |
| Contribution Margin (after variable costs) | (3%)  | 1%    | 4%         | 8%            |
| EBITDA Margin                              | (18%) | (12%) | (6%)       | 3%            |

The company turned contribution positive (revenue covering variable costs + customer acquisition) in FY25. However, it remains EBITDA-negative due to investments in:

- Dark store expansion (Rs.50-70 crore annually)
- Technology development (Rs.25-30 crore annually)
- Farmer training and sourcing infrastructure (Rs.10-15 crore annually)

Management targets EBITDA breakeven by FY28 as dark stores mature and customer acquisition costs decline through word-of-mouth.

## 4 Gap Analysis: Where NinjaCart Falls Short

Despite NinjaCart's impressive growth and healthy unit economics, significant gaps remain in its product portfolio, service delivery, technology, geographic coverage, and strategic positioning.

### 4.1 Product Portfolio Gaps

NinjaCart currently offers fresh fruits, vegetables, and staples. Missing categories include:

#### 4.1.1 1. Value-added produce:

- Cut vegetables (onion diced, tomato puree, mixed vegetable packs)
- Ready-to-cook (meal kits with pre-portioned vegetables + spices)
- Frozen vegetables (peas, corn, mixed vegetables)
- Fermented products (sauerkraut, kimchi, pickles - growing health segment)

**Market opportunity:** Rs.8,000 crore and growing at 30% CAGR. This is the fastest-growing segment, and NinjaCart has negligible presence.

#### 4.1.2 2. Exotic and specialty produce:

- Imported fruits (avocado, kiwi, blueberries, dragon fruit)
- Microgreens (broccoli, radish, sunflower - for high-end restaurants)
- Edible flowers (for fine dining and event catering)
- Organic certified produce (premium segment with 40%+ margins)

#### 4.1.3 3. Non-produce grocery:

- Dairy (milk, curd, paneer, cheese - high frequency, low margin)
- Staples (rice, dal, atta, oil - pantry fill, medium margin)
- Packaged snacks (biscuits, namkeen, ready-to-eat)

**Gap rationale:** Retailers who buy vegetables from NinjaCart must still source dairy, staples, and snacks from other distributors. This creates friction and limits wallet share.

#### 4.1.4 4. Farmer inputs (reverse integration):

- Seeds (high-quality, hybrid, disease-resistant)
- Fertilizers and pesticides (certified, genuine)
- Farm equipment (small tools, sprayers, harvesters)
- Crop advisory (weather-based, pest prediction)

**Gap rationale:** NinjaCart has sourcing relationships with 50,000+ farmers but does not sell inputs to them. This is a missed cross-sell opportunity and limits farmer loyalty.

## 4.2 Service Delivery Gaps

### 4.2.1 Geographic Concentration

Dark stores are concentrated in metropolitan areas and tier-1 cities:

Table 8: Dark Store Distribution by City

| City              | Stores    | Share       |
|-------------------|-----------|-------------|
| Bengaluru         | 12        | 27%         |
| Chennai           | 8         | 18%         |
| Hyderabad         | 6         | 13%         |
| Mumbai            | 5         | 11%         |
| Delhi NCR         | 5         | 11%         |
| Pune              | 3         | 7%          |
| Ahmedabad         | 2         | 4%          |
| Others (8 cities) | 4         | 9%          |
| <b>Total</b>      | <b>45</b> | <b>100%</b> |

**Significant gaps remain in:**

- Tier-2 cities (population 500,000-2,000,000) - 0 dark stores in cities like Madurai, Vijayawada, Bhopal, Chandigarh, Guwahati
- Tier-3 cities and rural retail - no presence
- Northern and Eastern India - underpenetrated compared to South

### 4.2.2 Delivery Time Performance

NinjaCart’s stated service level is 95% delivery by 7 AM. However, internal data shows:

Table 9: Delivery Performance by City

| City           | Delivery by 7 AM (%)      | Delivery by 8 AM (%) |
|----------------|---------------------------|----------------------|
| Bengaluru      | 89%                       | 96%                  |
| Chennai        | 91%                       | 97%                  |
| Hyderabad      | 87%                       | 94%                  |
| Mumbai         | 78% (traffic constraints) | 88%                  |
| Delhi NCR      | 75% (traffic, distance)   | 85%                  |
| Pune           | 88%                       | 95%                  |
| <b>Average</b> | <b>84%</b>                | <b>93%</b>           |

The bottom 10% of deliveries (those after 8:30 AM) generate 40% of all customer complaints—primarily from retailers whose stores open at 8 AM.

### 4.2.3 Quality Consistency Variability

While NinjaCart has standardized QC processes, quality varies significantly by source and season:

Table 10: Quality Performance by Category

| Quality Rating (retailer feedback)        | Percentage of Orders |
|-------------------------------------------|----------------------|
| 5-star (excellent)                        | 55%                  |
| 4-star (good)                             | 28%                  |
| 3-star (average)                          | 12%                  |
| 2-star or below (poor - complaints filed) | 5%                   |

The bottom 5% of orders (with quality complaints) generate 60% of customer churn. NinjaCart has not yet implemented a systematic process for source-level quality scoring to identify and eliminate consistently low-quality suppliers.

### 4.3 Technology Gaps

While NinjaCart’s technology stack is sophisticated relative to traditional supply chains, it lags behind best-in-class agri-tech platforms globally.

1. **No hyperlocal demand prediction:** Current demand forecasting is done at city and zone level, not at individual retailer or neighborhood level. This results in:
  - Stockouts on popular SKUs in some delivery routes
  - Spoilage on less popular SKUs in other routes
  - Inefficient allocation of inventory across dark stores

**Gap impact:** Estimated 5-8% of inventory value lost to misallocation.

2. **Limited farmer-facing technology:** The farmer mobile app lacks:
  - Real-time price discovery for different markets
  - Quality feedback (specific reasons for rejections)
  - Best planting advisory based on demand forecasts
  - Credit score and access to formal financing

**Gap impact:** Farmer retention is lower than retailer retention (only 40% of farmers sell consistently to NinjaCart beyond 3 seasons).

3. **No AI-based quality inspection:** Current QC is manual, subjective, and inconsistent. NinjaCart is piloting computer vision but has not deployed at scale. **Gap impact:** Quality disputes cost Rs.5-10 crore annually in refunds and lost retailer goodwill.
4. **No blockchain for traceability:** Consumers and regulators increasingly demand traceability (“where did this tomato come from?”). NinjaCart cannot currently provide farm-to-fork traceability. **Gap impact:** Missed premium pricing for traceable produce; regulatory risk as food safety norms tighten.
5. **No retailer credit scoring:** NinjaCart does not offer embedded credit. Retailers who cannot pay daily must either:
  - Borrow from informal sources at high interest rates

- Not order from NinjaCart (lost revenue)

**Gap impact:** Lost revenue from credit-constrained retailers. NinjaCart has the transaction data to build credit scores but has not done so.

#### 4.4 Strategic Gaps Relative to Key Competitors

A direct comparison with NinjaCart’s most formidable competitors illuminates strategic gaps.

Table 11: Strategic Comparison: NinjaCart vs. Competitors (1 = Poor, 5 = Excellent)

| Dimension                      | NinjaCart | WayCool | Jumbotail | Otipy |
|--------------------------------|-----------|---------|-----------|-------|
| Produce quality consistency    | 4         | 4       | 3         | 4     |
| Delivery reliability (on-time) | 4         | 4       | 3         | 4     |
| Product breadth (non-produce)  | 2         | 3       | 5         | 2     |
| Value-added products           | 1         | 3       | 2         | 2     |
| Farmer credit access           | 2         | 3       | 1         | 2     |
| Retailer credit access         | 3         | 3       | 4         | 2     |
| Technology sophistication      | 4         | 3       | 4         | 4     |
| Geographic coverage breadth    | 3         | 2       | 4         | 2     |
| Farmer trust (surveyed)        | 4         | 4       | 2         | 3     |
| Retailer trust (surveyed)      | 4         | 4       | 4         | 4     |

##### 4.4.1 Key Strategic Gaps Identified

1. **No credit offering:** WayCool offers input financing to farmers; Jumbotail offers working capital to retailers; Otipy has subscription-based model. NinjaCart has no formal credit product, losing credit-constrained customers on both sides.
2. **No farmer input linkage:** WayCool’s full-stack model captures farmers at input purchase, provides advisory, and then buys their output. A farmer who sells to NinjaCart may buy inputs from DeHaat or AgroStar. In that moment, NinjaCart loses cross-sell opportunity and farmer loyalty.
3. **Non-produce gap:** Jumbotail offers 3,000+ SKUs including staples, packaged foods, and personal care. A retailer using Jumbotail can source everything from one platform. NinjaCart’s produce-only focus means retailers must maintain multiple supplier relationships.
4. **Tier-2/3 presence:** Jumbotail has stronger presence in smaller cities where kirana stores are less serviced by modern B2B platforms. NinjaCart’s metro focus leaves this segment underpenetrated.
5. **Value-added gap:** As consumers shift toward convenience (cut vegetables, ready-to-cook), retailers will seek suppliers who offer these higher-margin products. NinjaCart’s current lack of these offerings may cause it to miss this wave.

## 5 Recommendations and Strategic Options

Based on the gap analysis, several strategic options emerge for NinjaCart's leadership. These are presented as discussion points for strategic planning.

### 5.1 Option 1: Deepen the Core Produce Business (Organic Growth)

Under this option, NinjaCart would double down on its core fresh produce model, accepting that it will not be a full-stack platform and instead focus on becoming the undisputed leader in B2B fresh produce supply in urban India.

#### 5.1.1 Key Strategies

##### 1. Expand dark stores to 100 locations by 2028, prioritizing:

- Tier-2 cities (Madurai, Vijayawada, Bhopal, Chandigarh, Guwahati, Lucknow, Jaipur) - 30 dark stores
- Northern India (Delhi NCR expansion, Chandigarh, Ludhiana, Agra) - 15 dark stores
- Eastern India (Kolkata, Bhubaneswar, Guwahati, Patna) - 10 dark stores

##### 2. Launch "NinjaCart Fresh Guarantee":

- 100% quality guarantee - full refund for any quality complaint
- 6 AM delivery guarantee - free order if late beyond 7 AM
- Builds retailer trust and reduces perceived risk

##### 3. Partner with NBFCs for retailer credit:

- 15-day payment terms for verified retailers (6+ months history)
- Dynamic credit limit based on order volume and repayment history
- NinjaCart takes 2-3% origination fee; NBFC takes interest

##### 4. Launch retailer loyalty program:

- Tier-based discounts (Bronze: 5% off on 10+ orders/month; Silver: 7% off; Gold: 10% off)
- Free delivery above certain order value
- Priority support for Gold tier

#### 5.1.2 Risks of Option 1

- **Competitive pressure:** WayCool and Jumbotail may slowly erode NinjaCart's retailer base by offering more complete solutions (non-produce + credit).
- **Capital intensity:** Expanding to 100 dark stores requires Rs.300-400 crore in capital expenditure over 3 years.
- **Slowing growth in core:** Fresh produce B2B growth is moderating (10-12% CAGR) compared to value-added (30% CAGR). Focusing only on produce may mean missing higher-growth, higher-margin segments.



## 5.2 Option 2: Build a Value-Added and Exotics Vertical

Under this option, NinjaCart would acquire or build a value-added processing capability, positioning itself for the high-growth convenience food market.

### 5.2.1 Key Strategies

#### 1. Acquire or build a central kitchen for cut vegetables:

- Estimated investment: Rs.50-100 crore
- Capacity: 10-15 tonnes/day of cut vegetables (onion, tomato, potato, mixed)
- Located in each major city near existing dark stores
- Pricing: Rs.60-100 per 500g pack (40% margin)

#### 2. Launch "NinjaCart Exotics" vertical:

- Source imported fruits (avocado, kiwi, dragon fruit) via direct import
- Partner with organic certified farmer clusters
- Target high-end retailers, fine dining restaurants, cloud kitchens
- Pricing: 30-50% premium over conventional produce

#### 3. Offer "Ready-to-Cook" meal kits:

- Pre-portioned vegetables + spices for popular dishes (paneer butter masala, palak paneer, mixed vegetable)
- Tie-up with spice brands (MDH, Everest, Catch)
- Target urban retailers in premium neighborhoods

#### 4. Integrate value-added into daily ordering:

- Add cut vegetables and meal kits to retailer order screen
- Cross-sell with fresh produce ("Add cut onions for Rs.30")
- Bundle discounts for produce + value-added

### 5.2.2 Risks of Option 2

- **Operational complexity:** Cut vegetables require additional food safety protocols (HACCP certification, temperature control, shelf-life management).
- **Capital intensity:** Central kitchens require significant upfront investment.
- **Competition from established players:** FreshToHome, Licious, and gourmet grocers already have presence in value-added.
- **Distraction from core:** Building a new vertical requires management attention that could otherwise go to scaling the core produce business.

### 5.3 Option 3: Integrate Backward into Farmer Inputs (Partnership Strategy)

Under this option, NinjaCart would partner with existing input platforms (DeHaat, AgroStar, BigHaat) rather than building competing capabilities.

#### 5.3.1 Key Strategies

##### 1. Formal partnership with DeHaat or AgroStar:

- NinjaCart farmers get 10% discount on inputs from partner
- Partner's farmers get priority access to NinjaCart's output procurement
- Revenue sharing on cross-sold customers (e.g., 5% of first-year input revenue)
- Joint marketing in shared geographies

##### 2. API integration for "input + output" bundles:

- Farmer buys seeds and fertilizers on partner app
- Checkout page offers "Sell harvest to NinjaCart at guaranteed price"
- Creates end-to-end farmer relationship without building both sides

##### 3. Data sharing agreement:

- Partner shares crop planting data (helps NinjaCart forecast harvest volumes)
- NinjaCart shares quality rejection data (helps partner improve seed/fertilizer recommendations)

##### 4. Joint farmer training programs:

- Partner provides agronomy expertise (crop planning, pest management)
- NinjaCart provides post-harvest training (handling, grading, packaging)
- Creates holistic farmer support without duplicating efforts

#### 5.3.2 Risks of Option 3

- **Dependence on partners:** DeHaat or AgroStar may decide to build their own output procurement capabilities (both have explored this).
- **Margin compression:** Discounts and revenue sharing reduce per-transaction margins.
- **Brand dilution:** Farmers may begin to see NinjaCart as "DeHaat's procurement arm" rather than an independent brand.
- **Data asymmetry:** Partner may capture more value from shared data than NinjaCart.

### 5.4 Option 4: Build a Retailer Full-Stack Platform (Kirana OS)

Under this option, NinjaCart would expand beyond produce to become an operating system for kirana stores—offering non-produce grocery, inventory management software, and embedded credit.

### 5.4.1 Key Strategies

#### 1. Launch "NinjaStore" - B2B grocery marketplace:

- Add 2,000+ SKUs of non-produce (staples, packaged foods, beverages, personal care, home care)
- Source from FMCG companies and distributors
- Take 8-12% commission on sales
- Cross-sell to existing produce retailers (low marginal acquisition cost)

#### 2. Acquire or build kirana POS/ inventory management software:

- Free or low-cost POS for kirana stores
- Automatic reordering based on sell-through data
- Inventory alerts (low stock, expiry management)
- Estimated development cost: Rs.10-20 crore

#### 3. Launch embedded credit for retailers:

- Working capital loans based on order history
- Repayment via daily sales deduction (seamless)
- Partnership with NBFCs (e.g., Avanti Finance, Kinara Capital)
- NinjaCart takes 2-3% origination fee

#### 4. Create kirana loyalty program across categories:

- Unified rewards for produce + grocery purchases
- Tier benefits (free delivery, priority support, credit access)
- Increases wallet share and reduces multi-supplier friction

### 5.4.2 Risks of Option 4

- **Competition with Jumbotail and Udaan:** Both have significant head start in non-produce B2B grocery.
- **Complexity:** Adding 2,000+ SKUs requires new sourcing, warehousing, and category management capabilities.
- **Capital requirements:** Inventory carrying costs for non-produce (slow-moving vs produce) tie up working capital.
- **Cannibalization risk:** Retailers may shift spend from produce to higher-margin non-produce, reducing overall profitability if margins are lower.

## 5.5 Option Comparison Matrix

Table 12: Strategic Options Comparison

| Criteria              | Produce      | Value-Added      | Partner         | Kirana OS        |
|-----------------------|--------------|------------------|-----------------|------------------|
| Capital (Rs. Cr)      | 300-400      | 50-100           | 10-20           | 100-200          |
| Time (months)         | 24-36        | 12-18            | 6-12            | 18-24            |
| FY30 Revenue (Rs. Cr) | 6k-7k        | 1k-1.5k          | 200-300*        | 5k-6k            |
| Margin                | Low (10-15%) | High (20-25%)    | Low (5-10%)     | Med (12-15%)     |
| Moat                  | Low          | Med              | Low             | High             |
| Distraction           | Low          | Med              | Low             | High             |
| Synergy               | High         | Med              | Med             | Med              |
| <b>Recommendation</b> | <b>Core</b>  | <b>Adjacency</b> | <b>Tactical</b> | <b>Long-term</b> |

## 6 Risk Assessment

### 6.1 Key Risks Facing NinjaCart

1. **Competition from deep-pocketed players:** WayCool has raised over \$100 million; Jumbotail is well-funded; Udaan could re-enter fresh produce aggressively. If any decides to price aggressively, NinjaCart could face margin pressure.
2. **Regulatory risk:** Changes in APMC mandi requirements, food safety regulations (FSSAI), or GST rates could increase compliance costs. For example, if the government mandates traceability for all fresh produce, NinjaCart would incur implementation costs that smaller competitors might avoid.
3. **Quality and perishability risk:** A single quality failure (e.g., contaminated produce, widespread spoilage) could damage retailer trust and trigger regulatory action. Unlike dry goods, fresh produce has inherent variability.
4. **Credit risk:** As NinjaCart begins offering credit (likely in the next 12-18 months), it will face default risk. Retailer credit scores are difficult to assess; many kirana stores have no formal financial history.
5. **Farmer concentration risk:** If a large farmer cluster (e.g., Kolar for tomatoes, Nashik for onions) faces crop failure, NinjaCart's sourcing for that SKU is compromised. Diversification across geographies mitigates but does not eliminate this risk.
6. **Talent risk:** NinjaCart competes with urban tech companies and e-commerce players for software engineers, data scientists, and product managers. Supply chain talent is scarce; the company may struggle to scale its technology team.

### 6.2 Risk Mitigation Strategies

Table 13: Risk Mitigation Matrix

| Risk                 | Mitigation                                                            |
|----------------------|-----------------------------------------------------------------------|
| Competition          | Quality, delivery reliability, retailer trust                         |
| Regulatory           | Proactive compliance + govt engagement (NABARD, Agri Ministry)        |
| Quality              | Enhanced QC, supplier scorecards, A-grade pricing                     |
| Credit risk          | Small tickets (Rs.5K-10K), short tenors (15-30 days), daily deduction |
| Farmer concentration | Multi-region sourcing, buffer inventory, spot mandi                   |
| Talent risk          | Remote center (tier-2 city), ESOPs                                    |

## 7 Conclusion: The Road Ahead

NinjaCart has successfully disrupted a traditional, inefficient market by solving a genuine retailer pain point: the combination of unreliable quality, fragmented sourcing, late deliveries, and no recourse for poor produce. The company's procurement-to-delivery flywheel, built on consistent quality, timely delivery, and retailer trust, has created real value for over 30,000 retailers and defensible unit economics (LTV:CAC of 8-10:1).

However, the gaps identified in this case—particularly the absence of value-added products, non-produce grocery, farmer credit, and tier-2/3 city presence—suggest that NinjaCart's current model may not be sufficient to achieve long-term dominance. The agri-tech landscape is evolving rapidly, with full-stack players (WayCool, DeHaat) and kirana platforms (Jumbotail) expanding into adjacent categories.

The strategic question facing NinjaCart's leadership is whether to:

- **Double down** on produce excellence (Option 1)
- **Expand into value-added and exotics** (Option 2)
- **Partner with input platforms** (Option 3)
- **Build a retailer full-stack platform (Kirana OS)** (Option 4)

There are compelling arguments on all sides. Option 1 is the safest and most aligned with current capabilities. Option 2 offers higher growth and margins but carries execution risk. Option 3 is the lowest cost but creates dependence on partners. Option 4 builds a durable moat but requires new competencies and significant capital.

For the small retailer, the ultimate measure of success remains simple: *increased profitability per square foot of store space*. NinjaCart has proven it can help on fresh produce—the highest-frequency, most-waste-prone category. The next frontier is total store fulfillment. The company's ability to navigate these strategic choices will determine whether it becomes a niche produce player or the operating system for India's 13 million kirana stores.

## Exhibits

### Exhibit A: NinjaCart Geographic Footprint (2026)

Table 14: NinjaCart Dark Store Distribution

| City         | Number of Dark Stores |
|--------------|-----------------------|
| Bengaluru    | 12                    |
| Chennai      | 8                     |
| Hyderabad    | 6                     |
| Mumbai       | 5                     |
| Delhi NCR    | 5                     |
| Pune         | 3                     |
| Ahmedabad    | 2                     |
| Lucknow      | 1                     |
| Jaipur       | 1                     |
| Coimbatore   | 1                     |
| Vizag        | 1                     |
| Nagpur       | 1                     |
| Indore       | 1                     |
| Surat        | 1                     |
| Kochi        | 1                     |
| <b>Total</b> | <b>45</b>             |

### Exhibit B: NinjaCart Growth Trajectory (FY22-FY28 Projected)

Table 15: Gross Merchandise Value (GMV) Growth

| Fiscal Year | GMV (Rs. Cr) |
|-------------|--------------|
| FY22        | 650          |
| FY23        | 1,100        |
| FY24        | 1,600        |
| FY25        | 2,100        |
| FY26        | 2,500        |
| FY27 (P)    | 3,100        |
| FY28 (P)    | 4,000        |

CAGR (FY22-FY26): 40%; Projected CAGR (FY26-FY28): 26%

**Exhibit C: Retailer Pain Point Summary**

Table 16: Top 10 Pain Points (% of surveyed retailers, N=2,500)

| <b>Pain Point</b>                                    | <b>% Respondents</b> |
|------------------------------------------------------|----------------------|
| 1. Inconsistent quality (variable freshness)         | 82%                  |
| 2. Delivery delays (affects store opening)           | 76%                  |
| 3. No recourse for poor quality (no refund/exchange) | 71%                  |
| 4. High prices compared to mandi                     | 65%                  |
| 5. Limited product selection (only staples)          | 59%                  |
| 6. Payment terms (upfront cash required)             | 54%                  |
| 7. No credit/working capital support                 | 48%                  |
| 8. Multiple suppliers needed (fragmented sourcing)   | 45%                  |
| 9. No digital ordering (phone/call only)             | 38%                  |
| 10. No loyalty benefits                              | 32%                  |

**Exhibit D: Fresh Produce Supply Chain Waste Analysis**

| <b>Post-Harvest Waste by Stage (Industry Average vs NinjaCart)</b> |                         |                     |
|--------------------------------------------------------------------|-------------------------|---------------------|
| <b>Stage</b>                                                       | <b>Industry Average</b> | <b>NinjaCart</b>    |
| Harvest to farm gate                                               | 5%                      | 3%                  |
| Farm gate to mandi                                                 | 10%                     | 4%                  |
| Mandi to wholesaler                                                | 5%                      | — (direct sourcing) |
| Wholesaler to retailer                                             | 5%                      | — (dark store)      |
| Retailer storage                                                   | 5%                      | 2%                  |
| <b>Total waste</b>                                                 | <b>30%</b>              | <b>9%</b>           |



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